Example :

$$Y_k = \mathbb{E}\{ \underbrace{f(\underline{X})} \mid \underbrace{X_1 \cdots X_k}_{\mathcal{F}_k = \sigma(X_1 \cdots X_k)} \}$$

Cond 1.

$$\mathbb{E}\{ Y_k \mid \mathcal{F}_{k-1} \} = Y_{k-1}$$

$$\mathbb{E}\{ \underbrace{\mathbb{E}\{ f(\underline{X}) \mid X_1 \cdots X_k \}} \mid X_1 \cdots X_{k-1} \} \overset{?}{=} \underline{\mathbb{E}}\{ f(\underline{X}) \mid X_1 \cdots X_{k-1} \}$$

$$= \iint f(\underline{x}) \cdot P(\overbrace{x_{k+1} \cdots x_n}^{A} \mid \overbrace{x_1 \cdots}^{C} \overbrace{x_k}^{B}) \, dx_{k+1} \cdots dx_n \cdot P(\overbrace{x_k}^{B} \mid \overbrace{x_1 \cdots x_{k-1}}^{C}) \, dx_k$$

$$P(A \mid C, B) \cdot P(B \mid C) = P(A, B \mid C)$$

$$= \int f(\underline{x}) \, P(x_k, x_{k+1}, \cdots, x_n \mid x_1 \cdots x_{k-1}) \, dx_k \, dx_{k+1} \cdots dx_n$$

$$= \mathbb{E}\{ f(\underline{X}) \mid X_1 \cdots X_{k-1} \}$$

Towering Property

Cond 2.

$$\mathbb{E}\{ |Y_k| \} < \infty$$

$$\mathbb{E}\{ | \mathbb{E}\{ f(\underline{X}) \mid X_1 \cdots X_k \} | \}$$

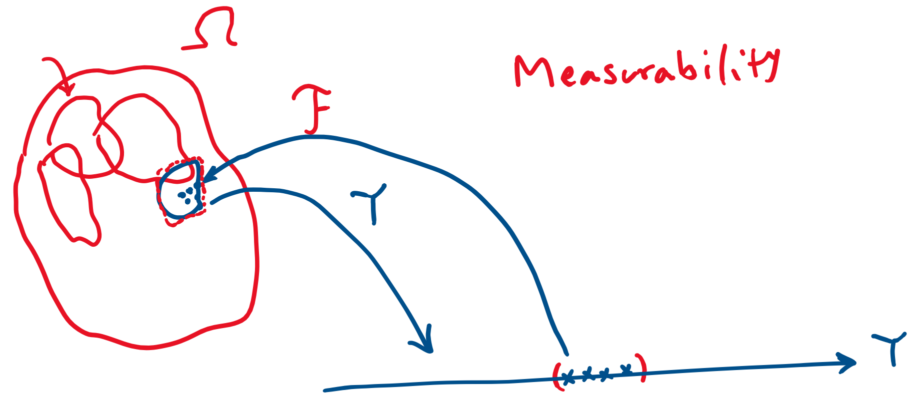
$$\leq \mathbb{E}\{ \mathbb{E}\{ |f(\underline{X})| \mid X_1 \cdots X_k \} \}$$

$$= \underbrace{\mathbb{E}\{ |f(\underline{X})| \} < \infty}_{\text{only if this is assumed}}$$

Guess:

$$\mathbb{E}\{ D_k \mid \mathcal{F}_{k-1} \} = 0 \quad \text{if } D_k \text{ is a Martingale Seq. Diff.}$$

$$\mathbb{E}\{ Y_k - Y_{k-1} \mid \mathcal{F}_{k-1} \} = \underbrace{\mathbb{E}\{ Y_k \mid \mathcal{F}_{k-1} \}}_{Y_{k-1}} - \underbrace{\mathbb{E}\{ Y_{k-1} \mid \mathcal{F}_{k-1} \}}_{Y_{k-1}} = 0$$



Measurability

Theorem: Assume that $\{D_k\}_{k=1}^{n}$ is Martingale Seq. Diff. ($\{Y_k, \mathcal{F}_k\}_{k=1}^{n}$ Mart.)

and $\underbrace{\forall_k \ \mathbb{E}\{e^{\lambda D_k} \mid \mathcal{F}_{k-1}\} \leq e^{\frac{1}{2}\nu_k^2\lambda^2}}_{\text{Big Assumption}}$ $\quad |\lambda| < 1/\alpha_k$.

Then,

$$\mathbb{P}\left(\left|\sum D_k\right| \geq t\right) \leq 2e^{-\frac{t^2}{2\sum_{k=1}^{n}\nu_k^2}}$$

$$0 < t < \frac{\sum \nu_k^2}{\alpha_*}$$

$$\alpha_* = \max_k \alpha_k$$

Proof: For this to happen, it suffices that $\sum D_k$ be subExponential.

$$\mathbb{E}\left\{e^{\lambda\left(\sum_{k=1}^{n} D_k - \mu\right)}\right\} \qquad \mu = \mathbb{E}\left\{\sum D_k\right\} = \sum \overbrace{\mathbb{E}(D_k)}^{0} = 0$$

$$\mathbb{E}(D_k) = \mathbb{E}\{\underbrace{\mathbb{E}\{D_k \mid \mathcal{F}_{k-1}\}}_{0}\} = 0$$

$$= \mathbb{E}\left\{e^{\lambda\sum_{k=1}^{n} D_k}\right\}$$

$$= \mathbb{E}\left\{\mathbb{E}\left\{\underbrace{e^{\lambda\sum_{k=1}^{n-1} D_k + \lambda D_n}}_{e^{\lambda\sum_{k=1}^{n-1} D_k} \cdot e^{\lambda D_n}} \mid \mathcal{F}_{n-1}\right\}\right\}$$

$$= \mathbb{E}\left\{e^{\lambda\sum_{k=1}^{n-1} D_k} \cdot \underbrace{\mathbb{E}\{e^{\lambda D_n} \mid \mathcal{F}_{n-1}\}}_{\leq e^{\frac{1}{2}\nu_n^2\lambda^2}}\right\} \leq e^{\frac{1}{2}\nu_n^2\lambda^2}\, \mathbb{E}\left(e^{\lambda\sum_{k=1}^{n-1} D_k}\right)$$

$$|\lambda| < \frac{1}{\alpha_n}$$

$$\leq e^{\frac{1}{2}\underbrace{\left(\sum \nu_k^2\right)}_{\nu^2}\lambda^2} \qquad |\lambda| < \frac{1}{\alpha_*} \qquad \alpha_* = \max_k \alpha_k$$

$$\Rightarrow \overbrace{\sum D_k}^{W} \sim \text{subE}\left(\nu^2 = \sum_{k=1}^{n} \nu_k^2,\ \alpha^* = \max \alpha_k\right)$$

$$\Rightarrow \mathbb{P}\left(\left|W - \mathbb{E}(W)\right| \geq t\right) \leq 2e^{\frac{1}{2}t^2/\nu^2} \quad 0 < t < \frac{\nu^2}{\alpha}$$

previously shown that

$\square$

The next step is to check whether or not the big assumption $\mathbb{E}\{e^{\lambda D_k} | \mathcal{F}_{k-1}\} \leq e^{\frac{1}{2}\nu_k^2\lambda^2}$ $|\lambda| < \frac{1}{\alpha_k}$ is satisfied, under what conditions.

Def. f is called Lipschitz under Hamming dist. if

$$\exists L \ \forall k \ \forall_{x_1 \cdots x_n\, x_k'} \left| f(x_1 \cdots x_k \cdots x_n) - f(x_1 \cdots x_k' \cdots x_n)\right| < L$$

Recall that if D_k is bounded, $D_k \in [a_k, b_k]$ a.s. $D_k \sim \text{subG}\left(\frac{b_k - a_k}{2}\right)$

$$B_k := \sup_x \mathbb{E}\{f(\underline{X}) | X_1 \cdots X_{k-1}\ X_k = x\} - \mathbb{E}\{f(\underline{X}) | X_1 \cdots X_{k-1}\}$$

$$B_k - D_k \overset{?}{\geq} 0 \text{ a.s.} \qquad D_k = \mathbb{E}\{f(\underline{X}) | X_1 \cdots X_{k-1}\ X_k\} - \mathbb{E}\{f(\underline{X}) | X_1 \cdots X_{k-1}\}$$

$$B_k - D_k = \sup_x \mathbb{E}\{f(\underline{X}) | X_1 \cdots X_{k-1}\ X_k = x\} - \mathbb{E}\{f(\underline{X}) | X_1 \cdots X_{k-1} X_k\}$$

$$\geq 0 \text{ a.s.}$$

$$A_k = \inf_{x} \mathbb{E}\{ f(\underline{X}) \mid X_1 \cdots X_{k-1}\ X_k = x \} - \mathbb{E}\{ f(\underline{X}) \mid X_1 \cdots X_{k-1} \}$$

$$D_k - A_k \geq 0 \quad \text{a.s.}$$

$$\Rightarrow \quad D_k \in [A_k, B_k] \quad \text{a.s.}$$

① $D_k \mid \mathcal{F}_{k-1} \in [A_k, B_k]$ a.s.

$$B_k - A_k = \sup_{x} \mathbb{E}\{ f(\underline{X}) \mid X_1 \cdots X_{k-1}\ X_k = x \}$$

$$- \inf_{x} \mathbb{E}\{ f(\underline{X}) \mid X_1 \cdots X_{k-1}\ X_k = x \}$$

$$= \mathbb{E}\{ f(\underline{X}) \mid X_1 \cdots X_{k-1}\ X_k = x^*_{max} \} -$$

$$\mathbb{E}\{ f(\underline{X}) \mid X_1 \cdots X_{k-1}\ X_k = x^*_{min} \}$$

indep of X_i's

$$= \mathbb{E}\{ f(X_1 \cdots X_{k-1}\ x^*_{max}\ X_{k+1} \cdots X_n) \}$$

$$- \mathbb{E}\{ f(X_1 \cdots X_{k-1}\ x^*_{min}\ X_{k+1} \cdots X_n) \}$$

$$= \mathbb{E}\{ f(X_1 \cdots X_{k-1}\ x^*_{max}\ X_{k+1} \cdots X_n) - f(X_1 \cdots X_{k-1}\ x^*_{min}\ X_{k+1} \cdots X_n) \}$$

$$\leq L. \qquad B_k - A_k \leq L.$$

$D_k \mid \mathcal{F}_{k-1} \sim \text{subG}\{L/2\}$

McDiarmid

$$\Rightarrow \mathbb{P}\left(f(X_1 \dots X_n) - \mathbb{E}\left(f(X_1 \dots X_n)\right) \geq t \right)$$

$$\leq e^{-t^2/(2nL^2/4)} = e^{-\frac{2t^2}{nL^2}}.$$